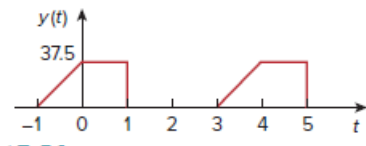


Problem

Obtain the exponential Fourier series for the signal in Fig. 17.54.

Figure 17.54:



Step-by-step solution

Step 1 of 4

Refer to signal in Figure 17.54 in the textbook.

The signal is,

$$f(t) = \begin{cases} 37.5 & 0 < t < 1 \\ 0 & 1 < t < 3 \\ 37.5(t-3) & 3 < t < 4 \end{cases}$$

A periodic function can be expressed in terms of exponential Fourier series is,

$$f(t) = \sum_{n=-\infty}^{\infty} c_n e^{jn\omega_0 t}$$

Here,

$$c_n = \frac{1}{T} \int_0^T f(t) e^{-jn\omega_0 t} dt$$

And,

$$\omega_0 = \frac{2\pi}{T} \text{ is the fundamental frequency}$$

Substitute 4 for T in the above equation.

$$\begin{aligned} \omega_0 &= \frac{2\pi}{4} \\ &= \frac{\pi}{2} \end{aligned}$$

Step 2 of 4

Calculate the coefficient c_n .

$$\begin{aligned}
 c_n &= \frac{1}{T} \int_0^T f(t) e^{-jn\omega_0 t} dt \\
 &= \frac{1}{4} \int_0^4 f(t) e^{-j\frac{n\pi t}{2}} dt \\
 &= \frac{1}{4} \left[\int_0^1 (37.5) e^{-j\frac{n\pi t}{2}} dt + \int_1^3 (0) e^{-j\frac{n\pi t}{2}} dt + \int_3^4 [37.5(t-3)] e^{-j\frac{n\pi t}{2}} dt \right] \\
 &= \frac{37.5}{4} \int_0^1 e^{-j\frac{n\pi t}{2}} dt + \frac{37.5}{4} \int_3^4 \left(te^{-j\frac{n\pi t}{2}} - 3e^{-j\frac{n\pi t}{2}} \right) dt \\
 &= \frac{37.5}{4} \left[\frac{e^{-j\frac{n\pi t}{2}}}{-j\frac{n\pi}{2}} \right]_0^1 + \frac{37.5}{4} \left[\frac{e^{-j\frac{n\pi t}{2}}}{\left(-j\frac{n\pi}{2}\right)^2} \left(-j\frac{n\pi t}{2} - 1\right) - 3 \frac{e^{-j\frac{n\pi t}{2}}}{-j\frac{n\pi}{2}} \right]_3^4 \\
 &= \frac{j75}{4n\pi} \left[e^{-j\frac{n\pi t}{2}} \right]_0^1 + \frac{j75}{4n\pi} \left[\frac{e^{-j\frac{n\pi t}{2}}}{\left(-j\frac{n\pi}{2}\right)^2} \left(-j\frac{n\pi t}{2} - 1\right) - 3e^{-j\frac{n\pi t}{2}} \right]_3^4 \\
 &= \frac{j75}{4n\pi} \left[e^{-j\frac{n\pi t}{2}} \right]_0^1 + \frac{j75}{4n\pi} \left[e^{-j\frac{n\pi t}{2}} \left(t - \frac{j2}{n\pi} \right) - 3e^{-j\frac{n\pi t}{2}} \right]_3^4 \\
 &= \frac{j75}{4n\pi} \left[e^{-j\frac{n\pi}{2}} - 1 \right] + \frac{j75}{4n\pi} \left[e^{-j2n\pi} \left(4 - \frac{j2}{n\pi} \right) - 3e^{-j2n\pi} - e^{-j\frac{n\pi 3}{2}} \left(3 - \frac{j2}{n\pi} \right) + 3e^{-j\frac{n\pi 3}{2}} \right]
 \end{aligned}$$

Step 3 of 4

On Further simplification:

$$\begin{aligned}
 c_n &= \frac{j75}{4n\pi} \left[e^{-j\frac{n\pi}{2}} - 1 + (1) \left(4 - \frac{j2}{n\pi} \right) - 3(1) - e^{-j\left(2n\pi - \frac{n\pi}{2}\right)} \left(3 - \frac{j2}{n\pi} \right) + 3e^{-j\left(2n\pi - \frac{n\pi}{2}\right)} \right] \\
 &= \frac{j75}{4n\pi} \left[e^{-j\frac{n\pi}{2}} - 1 + 4 - \frac{j2}{n\pi} - 3 - e^{j\frac{n\pi}{2}} \left(3 - \frac{j2}{n\pi} \right) + 3e^{j\frac{n\pi}{2}} \right] \\
 &= \frac{j75}{4n\pi} \left[e^{-j\frac{n\pi}{2}} - \frac{j2}{n\pi} - 3e^{j\frac{n\pi}{2}} + \frac{j2}{n\pi} e^{j\frac{n\pi}{2}} + 3e^{j\frac{n\pi}{2}} \right] \\
 &= \frac{j75}{4n\pi} \left[e^{-j\frac{n\pi}{2}} - \frac{j2}{n\pi} + \frac{j2}{n\pi} e^{j\frac{n\pi}{2}} \right] \\
 &= \frac{75}{4n^2\pi^2} \left[jn\pi e^{-j\frac{n\pi}{2}} + 2 - 2e^{j\frac{n\pi}{2}} \right] \\
 &= \frac{75}{4n^2\pi^2} \left[jn\pi \left[\cos\left(\frac{n\pi}{2}\right) - j\sin\left(\frac{n\pi}{2}\right) \right] + 2 - 2 \left[\cos\left(\frac{n\pi}{2}\right) + j\sin\left(\frac{n\pi}{2}\right) \right] \right] \\
 &= \frac{75}{4n^2\pi^2} \left[2 - 2\cos\left(\frac{n\pi}{2}\right) - j2\sin\left(\frac{n\pi}{2}\right) + jn\pi \cos\left(\frac{n\pi}{2}\right) + n\pi \sin\left(\frac{n\pi}{2}\right) \right]
 \end{aligned}$$

Step 4 of 4

Calculate the exponential Fourier series.

$$f(t) = \sum_{n=-\infty}^{\infty} c_n e^{jn\omega_0 t}$$

Substitute $\frac{75}{4n^2\pi^2} \left[2 - 2\cos\left(\frac{n\pi}{2}\right) - j2\sin\left(\frac{n\pi}{2}\right) + jn\pi\cos\left(\frac{n\pi}{2}\right) + n\pi\sin\left(\frac{n\pi}{2}\right) \right]$ for c_n in the above equation.

$$f(t) = \sum_{n=-\infty}^{\infty} \frac{75}{4n^2\pi^2} \left[2 - 2\cos\left(\frac{n\pi}{2}\right) - j2\sin\left(\frac{n\pi}{2}\right) + jn\pi\cos\left(\frac{n\pi}{2}\right) + n\pi\sin\left(\frac{n\pi}{2}\right) \right] e^{jn\omega_0 t}$$

Therefore, the exponential Fourier series, $f(t)$ is,

$$\boxed{\sum_{n=-\infty}^{\infty} \frac{75}{4n^2\pi^2} \left[2 - 2\cos\left(\frac{n\pi}{2}\right) - j2\sin\left(\frac{n\pi}{2}\right) + jn\pi\cos\left(\frac{n\pi}{2}\right) + n\pi\sin\left(\frac{n\pi}{2}\right) \right] e^{jn\omega_0 t}}.$$